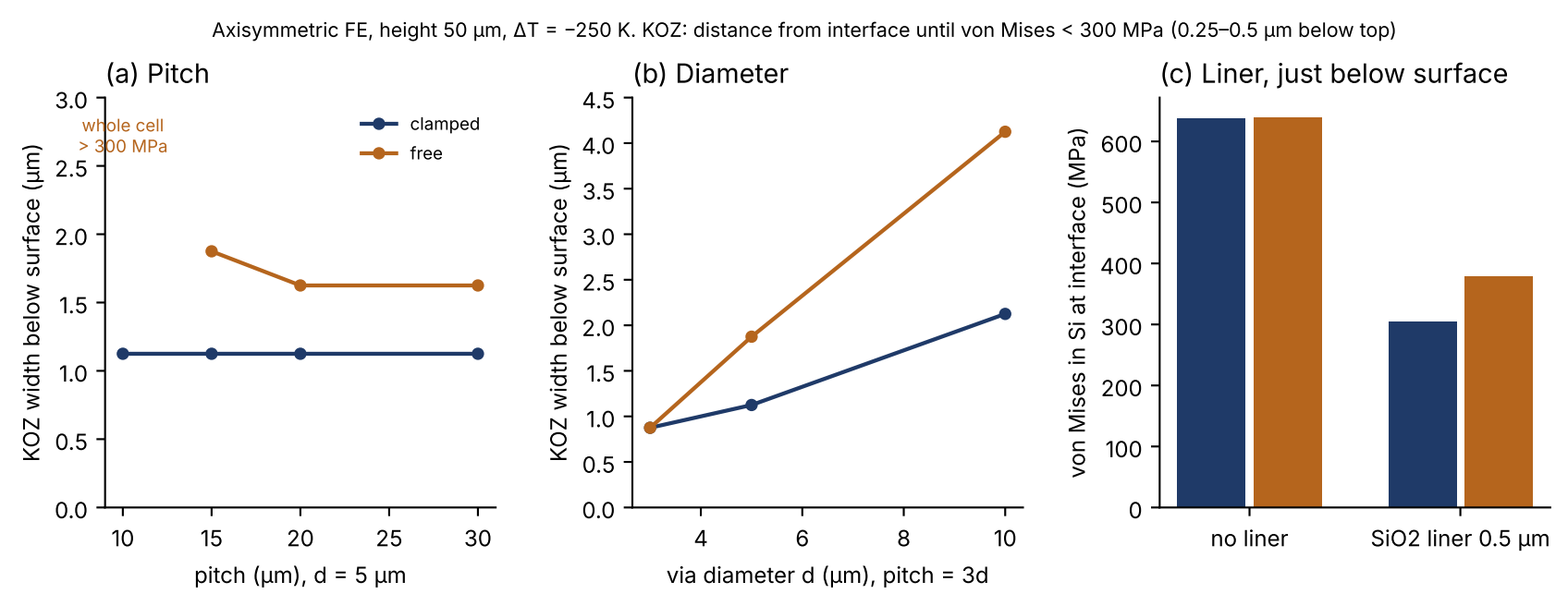
Axisymmetric FE, height 50 µm, ΔT = −250 K. KOZ: distance from interface until von Mises < 300 MPa (0.25–0.5 µm below top)
(a) Pitch
whole cell
> 300 MPa
clamped
free
KOZ width below surface (µm)
pitch (µm), d = 5 µm
(b) Diameter
KOZ width below surface (µm)
via diameter d (µm), pitch = 3d
(c) Liner, just below surface
von Mises in Si at interface (MPa)
no liner
SiO2 liner 0.5 µm